

1. INPUT SPECIFICATIONS

Required capacity	100.0 t/h
Lift height	25.0 m
Material	Clinker
Bulk density	1500 kg/m3
Service environment	Dry
Site wind pressure	800 Pa

2. PERFORMANCE SUMMARY

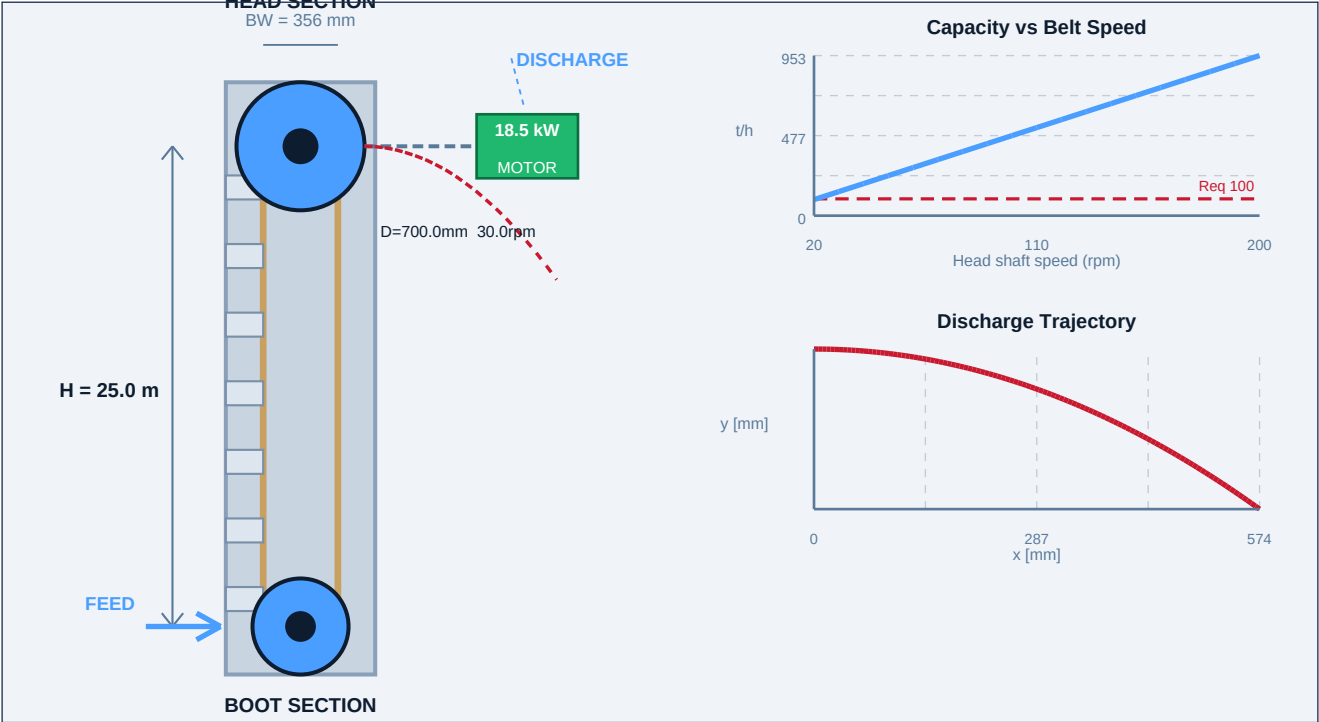
Capacity 143.0 t/h	Belt Speed 1.10 m/s	Total Power 14.3 kW	Motor Selected 18.5 kW	Chain SF 8.95 —	Bearing L10 329,220 h
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3. PROCESS LEVEL OUTPUTS

Design Configuration

Head pulley dia.	700.0 mm	Belt type	EP
Head shaft speed	30.0 rpm	Belt friction mu	—
Boot pulley dia.	300.0 mm	Wrap angle	— deg
Fill factor	64%	Service factor	—

System Schematic & Performance Charts



Achieved capacity	143.0 t/h	Discharge angle	0.0 deg from vertical
Chain speed	1.10 m/s	Belt width	356 mm
Bucket spacing	0.228 m	Recommended fill	64.0%

Centrifugal ratio	0.352	Stream spread	0.115
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4. POWER BREAKDOWN

Boot equivalent height (D_boot x Leq)	5.60 m
Total equivalent height (H + H_equiv)	30.60 m
Lift power (material only)	9.74 kW
Boot scooping power (boot loading)	2.18 kW
Shaft power (total load at head shaft)	11.92 kW
Drive losses (gearbox + bearings + belt flex)	2.38 kW
Total motor power (shaft x Ceff)	14.31 kW
Design power (x service factor 1.25)	17.89 kW
Selected motor	18.5 kW
Gearbox ratio (at 1450 rpm input)	48.3 : 1
Drive efficiency factor Ceff	1.200
CEMA LEQ factor	8.0

5. COMPONENT DESIGN OUTPUTS

5a. Head Shaft & Drive

Material grade	ASTM A36 Mild Steel	Hub connection	Keyed (ASME B17.1)
Shaft torque	4554.7 Nm	Hub OD	168.4 mm
Section	Solid	Hub length	140.4 mm
Min shaft dia.	112.3 mm	Key b x h	28 x 16 mm
Governed by	stress	Key shear	20.64 / 42.0 MPa
Allowable shear	42 MPa (keyed)	Key bearing	72.23 / 215.0 MPa
Bearing span	686 mm	Key result	PASS
Drive arm A	309 mm	Bearing L10	329220 h
Tail arm B	377 mm		

5b. Belt & Bucket Selection

Bucket series	AC_12x8	Belt width	356 mm
Style	CC	Belt class	1 PLY
Width x depth	305 x 216 mm	Belt length (total)	53.7 m (incl. splice allowance)
Projection	203 mm	Bucket spacing (actual)	0.228 m
Volume (struck)	8.58 L	Bucket count	229 off
Bucket mass	11.00 kg	Fill factor	64%
		Material DB fill	64.0% advisory

5c. Pulley Design

Lagging type	—	Shell min t (CEMA)	13.0 mm
Lagging thickness	— mm	Shell min t (press)	0.65 mm
mu dry / wet	— / —	Shell governing t	13.0 mm (CEMA minimum)
mu operating	—	Shell specified t	13.0 mm (= calculated minimum, no override)
Euler limit	—	End disc min t	28.2 mm
Belt ratio R/T3	—	End disc specify	33 mm (+20%)
Slip check	—	Disc governed by	plate_bending

		Disc sigma bend	80.0 MPa
		Force per disc	21.13 kN
		Shaft critical speed	2237 rpm
		Operating ratio	30 rpm (1% of critical)

5d. Take-Up Design

Type	Gravity (primary)	Alt type	Screw (alternative)
Counterweight net	3548.0 kg	Screw load	40.95 kN
Counterweight gross	3902.8 kg	Min core dia.	29.5 mm
Travel required	470 mm	Turns required	78.0
— thermal	48 mm	Buckling SF	4.96
— elongation	122 mm	Buckling check	PASS

5e. Casing & Structural

Plate thickness	8.2 mm	Panel delta actual	2.99 mm
Max stiffener pitch	2065.0 mm	Panel delta allowed	4.88 mm
Recommended pitch	1755.0 mm	Panel sigma max	11.1 MPa
Deflection limit	L / 360	Panel check	PASS
Wind pressure	800.0 Pa	Bolt fatigue Goodman	0.001
		Assembly fasteners	M12 x 793 (panel-to-panel + stiffener)

5f. Discharge Chute Design

Back-plate angle	67.8 deg	Spout width	405.0 mm
Min angle (wall fric.)	40.0 deg	Throw distance	0.29 m
Angle adequate	YES	Throat velocity	0.416 m/s
Flow regime	MASS_FLOW	Geom plugging risk	HIGH
Mass-flow threshold	31.5 deg	Hood radius	0.438 m
Governed by	angle of repose	Capture efficiency	0.75
Wear index	6.05 (MEDIUM)	Plugging risk	HIGH
Liner material	AR400 Wear Plate	Plugging index	1.205
Liner thickness	10 mm	Dust risk	MEDIUM
Liner grade	400 BHN	Recommended sensors	flow_sensor_acoustic , chute_blockage_detector

Chute Recommendations

* Plugging index 1.21 (HIGH) — Air cannons or fluidising pads; steep chute angle required.

* Wear index 6.0 (MEDIUM) — AR400 Wear Plate liner recommended.

5g. Feed Design (Boot Section)

Loading method	Centrifugal – Digging / Scooping	Material depth	152.0 mm (0.75xP)
Volumetric flow	95.34 m ³ /h	Dig zone arc	550.0 mm
Inlet velocity	0.3 m/s	Dig volume	29.8 L
Inlet area req.	882.7 cm ²	Surge volume	79.4 L (3.0s)
Inlet width	356.0 mm	Boot casing h	819.0 mm
Inlet height	152.0 mm	Floor clearance	50.0 mm

5h. Chain Drive Configuration

Chain series	ER-857 (6" MDC rollerless, single)	Chain pull	24.86 kN
Chain pitch	152.4 mm	SF actual	8.95
No. of strands	1	Speed check	PASS
Working load	22680 kg / strand	Sprocket teeth	14
Chain weight	8.5 kg/m	Sprocket PD	684.9 mm
Max chain speed	1.35 m/s	Smooth op.	YES

6. ENGINEERING VERIFICATION

Check	Actual	Limit	SF	Result
Capacity	143.0 t/h	100.0 t/h	1.43	✓ OK
Chain working load SF	8.95	>= 6.0 (CEMA 375)	8.95	✓ OK
Chain speed $v \leq$ rated	1.10 m/s	1.35 m/s rated	1.23	✓ OK
Sprocket 10–20 teeth (smooth)	14 teeth	10 – 20	–	✓ OK
Headshaft radial load	42.26 kN	80 kN	1.89	✓ OK
Shaft diameter (stress)	112.3 mm calc	112.3 mm governing	1.00	i INFO
Bearing L10 life	329,220 h	20,000 h min	16.46	✓ OK
Keyway shear stress	20.64 MPa	42.0 MPa	2.03	✓ OK
Keyway bearing stress	72.23 MPa	215.0 MPa	2.98	✓ OK
Bolt fatigue Goodman ratio	0.001	< 1.000 (infinite life)	1000.00	✓ OK
Casing panel deflection	2.99 mm	4.88 mm (L/360)	1.63	✓ OK
Casing clearance (stream vs wall)	573.6 mm stream reach	228.0 mm wall	-345.6 mm margin	✗ FAIL
Stream → chute interception	28.4 deg impact angle	2.28 m/s impact velocity	–	✓ OK

Corrective Actions

■ PLUGGING — Chute plugging probability HIGH despite adequate angle — cohesive material

1. Specify AR400 Wear Plate liner to reduce surface adhesion
2. Install vibration motor or air cannon on chute back-plate
3. Consider chute geometry with steeper secondary slope

7. DESIGN NOTES

Capacity & Selection: The elevator is configured with Series AC_12x8 buckets and achieves a throughput of 143.0 t/h against the 100 t/h requirement, providing a capacity margin of 43%. Belt speed is 1.10 m/s at the specified head shaft speed, which is within the acceptable range for the selected bucket geometry. The centrifugal discharge ratio of 0.352 is outside the ideal centrifugal range and should be reviewed.

Head Shaft & Hub: The head shaft minimum diameter is 112 mm, governed by stress. The shaft spans 686.0 mm between bearing centrelines. The hub outer diameter is 168.4 mm with a hub engagement length of 140.4 mm. The keyway stress check passes at the selected shaft and hub dimensions.

Pulley Lagging: Rubber herringbone lagging, — mm thick, is specified for the head pulley. This provides a belt-to-pulley traction coefficient of — under dry service conditions. Belt slip analysis confirms the lagging selection is adequate at operating tensions.

Pulley End Disc: Pulley end discs require a structural minimum thickness of 28.2 mm (governed by plate_bending). A specified thickness of 33 mm is recommended in fabrication drawings to provide adequate construction tolerance and weld preparation allowance. Full finite element analysis or detailed annular plate calculation is recommended prior to fabrication.

Take-Up System: The gravity take-up counterweight gross mass is 3902.8 kg. The take-up frame must accommodate a minimum carriage travel of 470 mm, combining thermal expansion, belt elastic elongation, and installation clearance components. A 20% allowance above this minimum is recommended for field adjustment headroom. As an alternative for short elevators, a screw take-up requires a minimum screw core diameter of 29.5 mm. The screw buckling safety factor is adequate.

Casing & Structural: The casing plate thickness of 8.2 mm is determined by material bulk density and elevator height. Structural stiffeners at 1755.0 mm centres limit panel deflection to within acceptable serviceability limits under the 800.0 Pa design wind load. Panel deflection analysis confirms the stiffener layout is adequate.

Power & Motor: The calculated shaft power is 11.924 kW. Total motor input power including drive losses is 14.3 kW. A 18.5 kW motor has been selected, representing a 29% power margin above the design requirement at the specified service factor. The gearbox reduction ratio should be confirmed with the selected motor supplier based on motor synchronous speed.

Bearing Life: Head shaft bearing life is calculated at 329,220 hours, which exceeds standard requirements. Bearing selection should be confirmed from the manufacturer catalogue using the equivalent dynamic load at the specified shaft speed. Lubrication interval and replacement schedule should be established based on the supplier's rated life data for the selected bearing series.

Bucket Bolt Fatigue: Bucket mounting bolt fatigue has been assessed using the Goodman diagram method. The fatigue utilization ratio is 0.001, which is well within the infinite life threshold — bolt fatigue is not a design concern at the specified operating conditions.

Discharge Chute: The discharge chute back-plate angle of 67.8 degrees is set by the material trajectory. Flow analysis indicates mass flow conditions at the specified belt speed. AR400 Wear Plate liner, 10 mm thick, is specified based on the calculated wear index. Plugging probability is high and dust generation risk is medium for this material. The material throw distance is 0.29 m from the pulley centreline, which should be verified against the physical head section geometry.

Feed Design (Boot Section): The elevator uses centrifugal digging to fill buckets at the boot. The material depth in the boot pit should be maintained at approximately 152.0 mm (0.75 times the bucket projection) for consistent scooping. The active digging zone spans 550.0 mm of arc at the boot pulley, with an estimated digging volume of 29.8 litres. The boot surge volume is 79.4 L (3.0-second buffer). The boot casing floor should be a minimum of 819.0 mm below the boot pulley centreline to provide adequate clearance.

8. ENGINEERING SIGN-OFF

This report has been prepared in accordance with CEMA No. 375-2017 and ANSI/CEMA 550-2020. The signatory confirms that the design calculations have been reviewed and are accepted as the basis for procurement and fabrication.

DESIGNED BY Name: Designation / Company: Date: Signature STAMP / SEAL	REVIEWED BY Name: Designation / Company: Date: Signature STAMP / SEAL	APPROVED BY Name: Designation / Company: Date: Signature STAMP / SEAL
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Standards: CEMA No. 375-2017 · ANSI/CEMA 550-2020 · ISO 281 · ASME B17.1 · AS 4024 (where applicable)